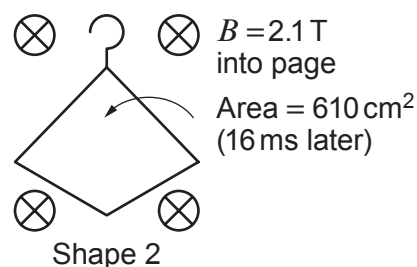
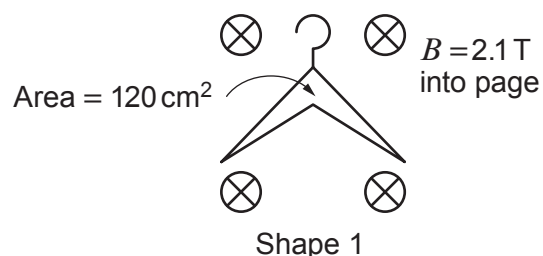


5. An experiment is carried out in a very strong uniform magnetic field in order to confirm Faraday's Law under extreme conditions. A coat hanger made of aluminium wire is bent from Shape 1 to Shape 2 in a time of 16 ms.



- (a) (i) Show that a mean emf of 6.4 V is induced in the coat hanger. [2]

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- (ii) Show on the diagram of Shape 2 the direction of the induced current and state very briefly how you determined this direction. [2]

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- (b) The aluminium wire has a circular cross-section of diameter 3.0 mm and the length of the wire through which the current flows is 91 cm. Show that the mean current in the wire is approximately 1 900 A (resistivity of aluminium = $2.65 \times 10^{-8} \Omega \text{ m}$). [3]

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- (c) lestyn claims that changing the shape of the coat hanger from Shape 1 to Shape 2 in 16 ms in the magnetic field will increase its temperature by less than 1°C . Determine, using appropriate calculations, whether or not lestyn is correct. (Density of aluminium = 2700 kg m^{-3} , specific heat capacity of aluminium = $897\text{ J kg}^{-1}\text{ K}^{-1}$.) [5]

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- (d) For medical research, it is decided to investigate the effect of this strong magnetic field (2.1 T) on patients with metal replacement joints to see if the metal joints become hot or undergo large forces (during MRI scans). Discuss the ethics of such an experiment. [3]

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